PUMP & MOTOR CONDITION MONITORING REPORT

Intelligent Predictive Maintenance Framework · Smart Manufacturing Decision Support

IMPORTANT NOTICE: ACADEMIC ENGINEERING PROTOTYPE ONLY.
This diagnostic report is generated from local simulations and sensor features for educational and decision-support research. It is NOT certified for physical plant protection or automated safety trips.

1. MACHINE TECHNICAL PROFILE & APPLICABLE CRITERIA

Machine ID: M1213 | Equipment Name: Chemical Circulation Pump
ISO 10816-7 Classification: Category 1 | Nominal Power: 280.0 kW
Rated Speed: 3000 RPM (Running Freq: 50.0 Hz) | Standard Limits: Europump 2013 Table 1
Report Generation Timestamp: 2026-10-26 16:50:21 UTC

2. RECENT CONDITION ASSESSMENTS & DIAGNOSTIC EVENTS

Event #5291 · Date: 2025-03-02 | Status: OPEN | Severity: Zone D (Hazardous / Damage risk) (Peak: 8.69 mm/s RMS)
Candidate Fault: Rotor Imbalance (Confidence Score: 0.55)

Event #5290 · Date: 2025-03-01 | Status: OPEN | Severity: Zone D (Hazardous / Damage risk) (Peak: 7.98 mm/s RMS)
Candidate Fault: Unknown / Insufficient Evidence · Physical Inspection Required (Confidence Score: 0.35)

Event #5289 · Date: 2025-02-28 | Status: OPEN | Severity: Zone D (Hazardous / Damage risk) (Peak: 7.99 mm/s RMS)
Candidate Fault: Unknown / Insufficient Evidence · Physical Inspection Required (Confidence Score: 0.35)

Event #5288 · Date: 2025-02-27 | Status: OPEN | Severity: Zone D (Hazardous / Damage risk) (Peak: 8.54 mm/s RMS)
Candidate Fault: Unknown / Insufficient Evidence · Physical Inspection Required (Confidence Score: 0.35)

3. RECOMMENDED MAINTENANCE ACTIONS & SIGN-OFF

1. Physical Verification: Verify transducer stud torque and confirm no loose accelerometer cables.
2. Handheld Walkdown: Conduct spectral frequency survey around 1x RPM and coupling axial plane.
3. Lubrication Inspection: Check oil reservoir level and verify grease interval compliance.
4. Work Order Authorization: Operator must verify findings prior to opening physical mechanical casing.